

Analysis of Existing Wildfire Detection Systems and Proposed Novel Framework

1. Existing Features in the 12 Papers

Fire Detection and Mapping Features

- The reviewed papers collectively focus on active fire detection using thermal anomalies, burned area mapping, fire intensity classification, pixel-wise fire segmentation, and ignition point localization using smoke analysis. These approaches enable identification of fire-affected regions and support disaster assessment and forensic investigation. However, most methods detect fires only after ignition or damage and are not optimized for early warning using near real-time (NRT) satellite data.

Satellite and Data Usage

- The papers utilize satellite sources such as MODIS, VIIRS, Sentinel-2, and SAR data, leveraging multispectral bands including SWIR, NIR, and thermal channels. These data sources provide reliable large-scale geographical coverage. Nonetheless, most approaches rely heavily on image-based data, whereas the proposed project dataset is event-based tabular NRT fire data.

Modeling Techniques

- Existing works employ convolutional neural networks, generative adversarial networks, transformers, anomaly indices, and physics-inspired divergence models. While these techniques achieve high accuracy, they are computationally intensive, less suitable for real-time deployment, and often function as black-box models with limited interpretability.

Temporal Analysis

- Several papers incorporate multi-temporal satellite imagery and time-series indices to capture fire evolution. However, none effectively model event-level temporal intelligence or decision escalation for early alerts.

Output and Decision Support

- Outputs include fire detection maps, intensity levels, and burned area assessments. These are useful for analysis but lack actionable alerting mechanisms, confidence scoring, or readiness indicators.

Advantages of Existing Systems

- The existing systems demonstrate strong detection accuracy, leverage proven satellite platforms, and support post-fire assessment and risk analysis. They provide valuable insights for environmental monitoring but are limited in real-time operational applicability.

Identified Research Gap

- None of the twelve papers fully exploit near real-time fire event streams such as NASA FIRMS for early-warning decision support. Missing aspects include event-based temporal modeling, confidence-aware alerting, persistence analysis, and operational action recommendations.

Proposed Novel Framework (Your research paper)

- The proposed project introduces an event-based early wildfire risk assessment framework using only the NASA FIRMS Near Real-Time dataset. Instead of image-based detection, the system analyzes fire event streams characterized by location, timestamp, brightness temperature, fire radiative power, and confidence scores.
- Novel features include:
 - Fire persistence analysis to differentiate false positives from genuine ignition events
 - Temporal escalation modeling using trends in fire radiative power and brightness
 - A Fire Alert Readiness Index (FARI) that classifies events into actionable risk levels
 - Explainable alerts that clearly indicate why an alert was triggered

Paper Title:

Spatiotemporal Modeling of Near Real-Time Satellite Fire Events for Early Wildfire Risk Assessment

IEEE-Style Abstract (150 words):

- Early detection of wildfires is critical for minimizing environmental damage and enabling rapid emergency response. While existing wildfire detection systems rely heavily on satellite imagery and deep learning models, they often lack real-time decision-making capability and operational interpretability. This paper proposes a novel event-based early wildfire risk assessment framework using NASA's Near Real-Time Fire Information for Resource Management System (FIRMS) dataset. The proposed system analyzes spatiotemporal fire event streams by modeling fire persistence, intensity escalation, and spatial clustering using brightness temperature, fire radiative power, detection confidence, and temporal recurrence. A new Fire Alert Readiness Index (FARI) is introduced to classify detected fire events into actionable risk levels, enabling authorities to distinguish between transient thermal anomalies

and potential wildfire ignitions. Unlike image-based approaches, the proposed framework operates directly on near real-time satellite fire data, ensuring computational efficiency, scalability, and explainability. Experimental analysis demonstrates the system's effectiveness in providing timely, interpretable early-warning alerts suitable for real-world wildfire monitoring and decision support.

Advantages of the Proposed Project:

1. Technical Advantages:
 - Utilizes real-time satellite fire event data
 - Eliminates dependency on heavy image processing pipelines
 - Scalable to regional and global monitoring
 - Low computational complexity
2. Operational Advantages:
 - Enables early warning instead of post-fire assessment
 - Provides actionable alert categories
 - Confidence-aware decision support for authorities
3. Research Advantages:
 - Clear novelty over image-based wildfire detection systems
 - Dataset-driven and reproducible methodology
 - Strong alignment with IEEE publication standards